

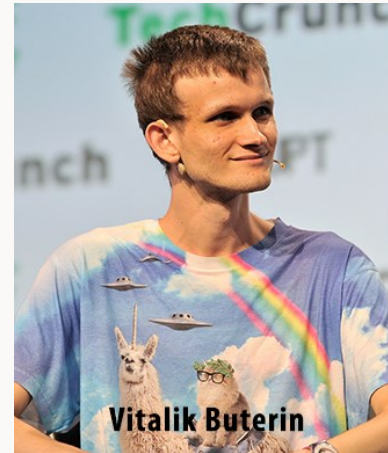


ethereum

Blockchain Principles and Applications
Sharif University of Technology
Spring 2025

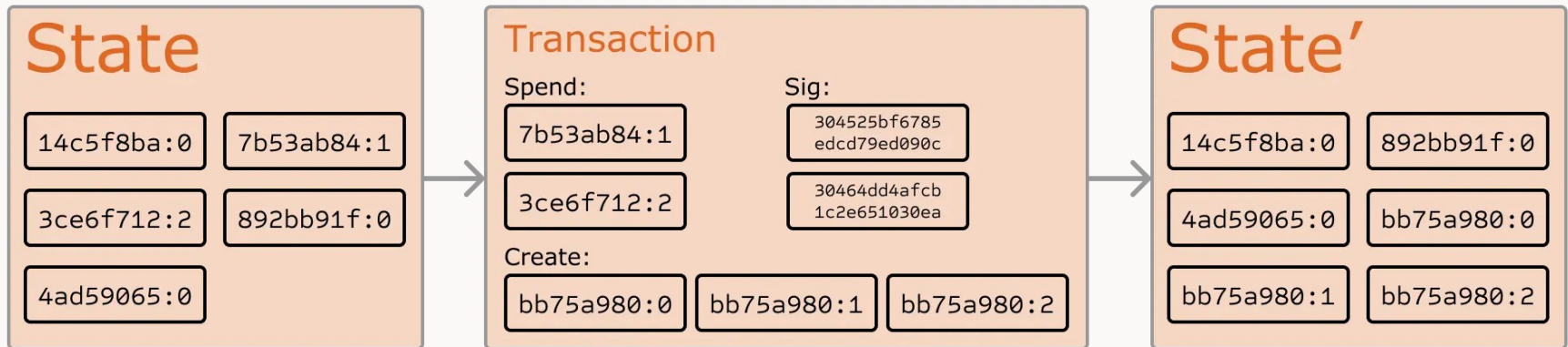
Background

- From Aether /'i:θər/
 - fifth element
- Whitepaper in 2013
 - by Vitalik Buterin
- Yellow paper in 2014
 - by Gavin Wood
- *Ethereum Foundation* founded in 2014
- Public network launched in 2015



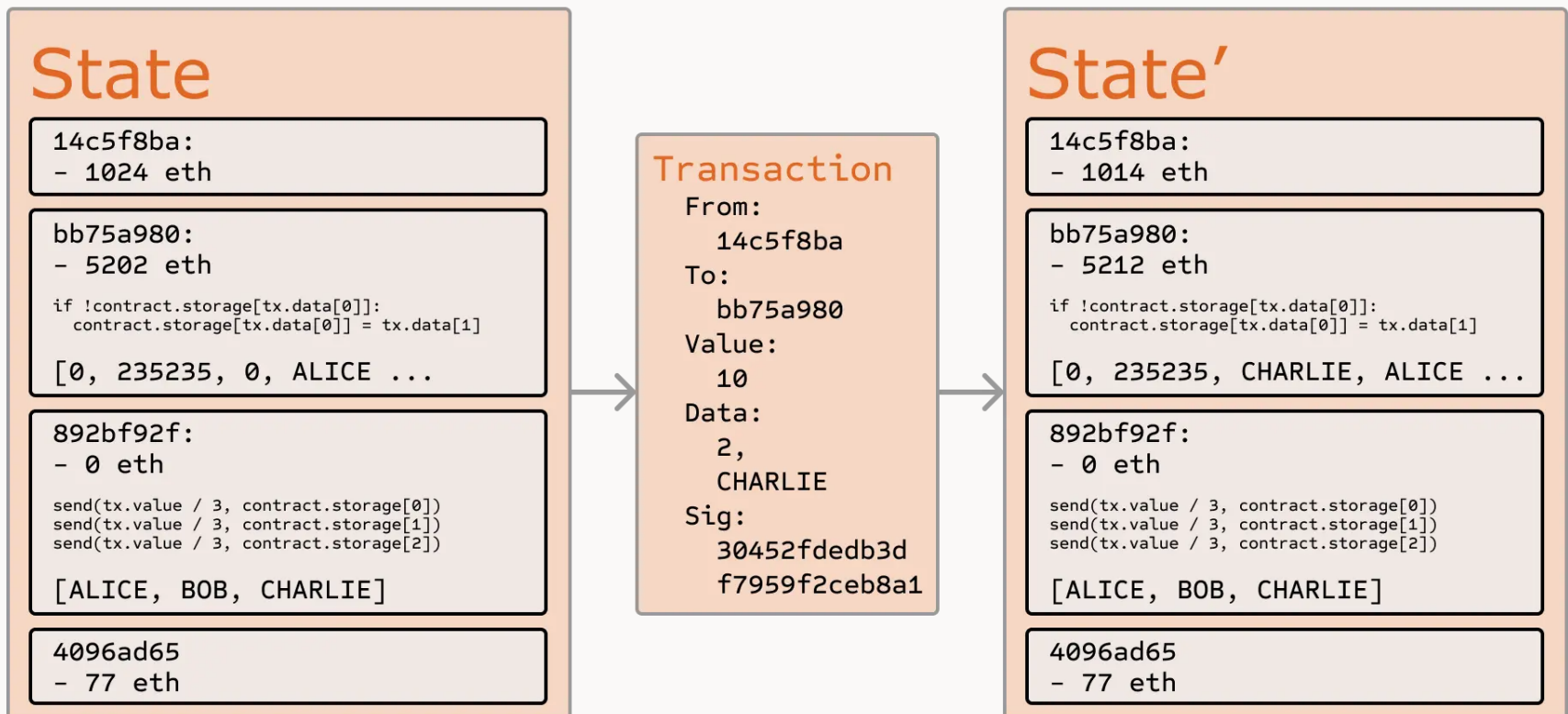
What is Ethereum ?

- State Transition System
 - use account instead of UTxO
- Bitcoin State : ID



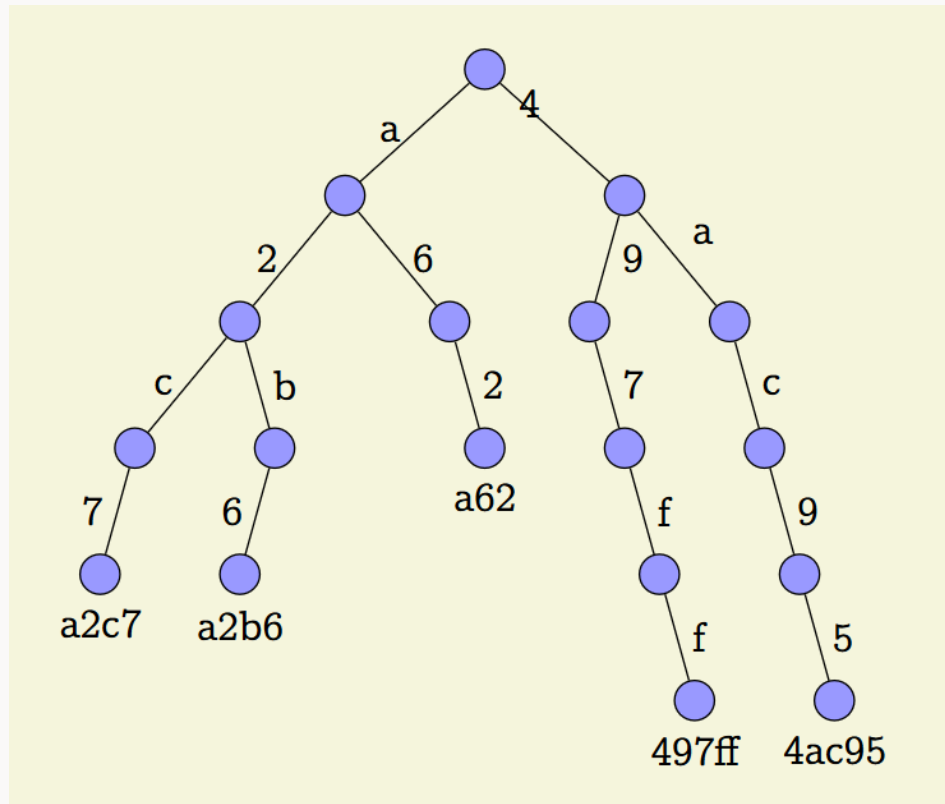
What is Ethereum ?

- State Transition System
- Ethereum State: ID+Balance+Storage+Nonce



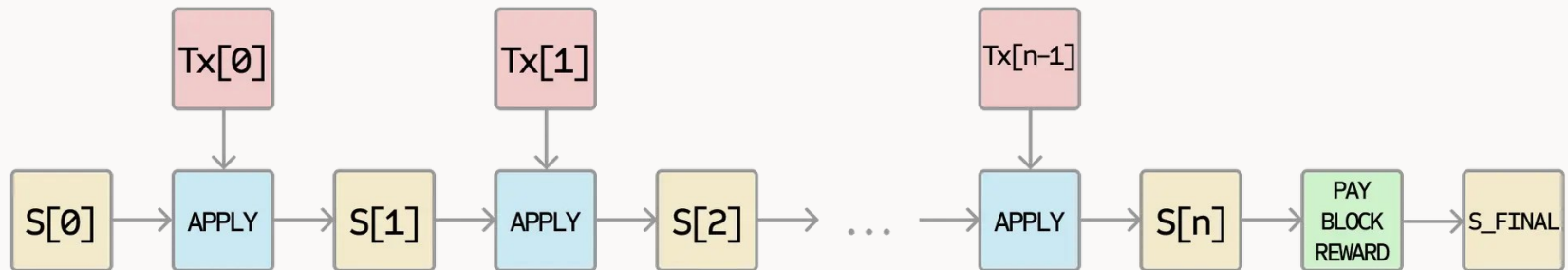
What is Ethereum ?

- State Transition System
- Ethereum State: ID+Balance+Storage+Nonce
- Merkle-Patricia trie



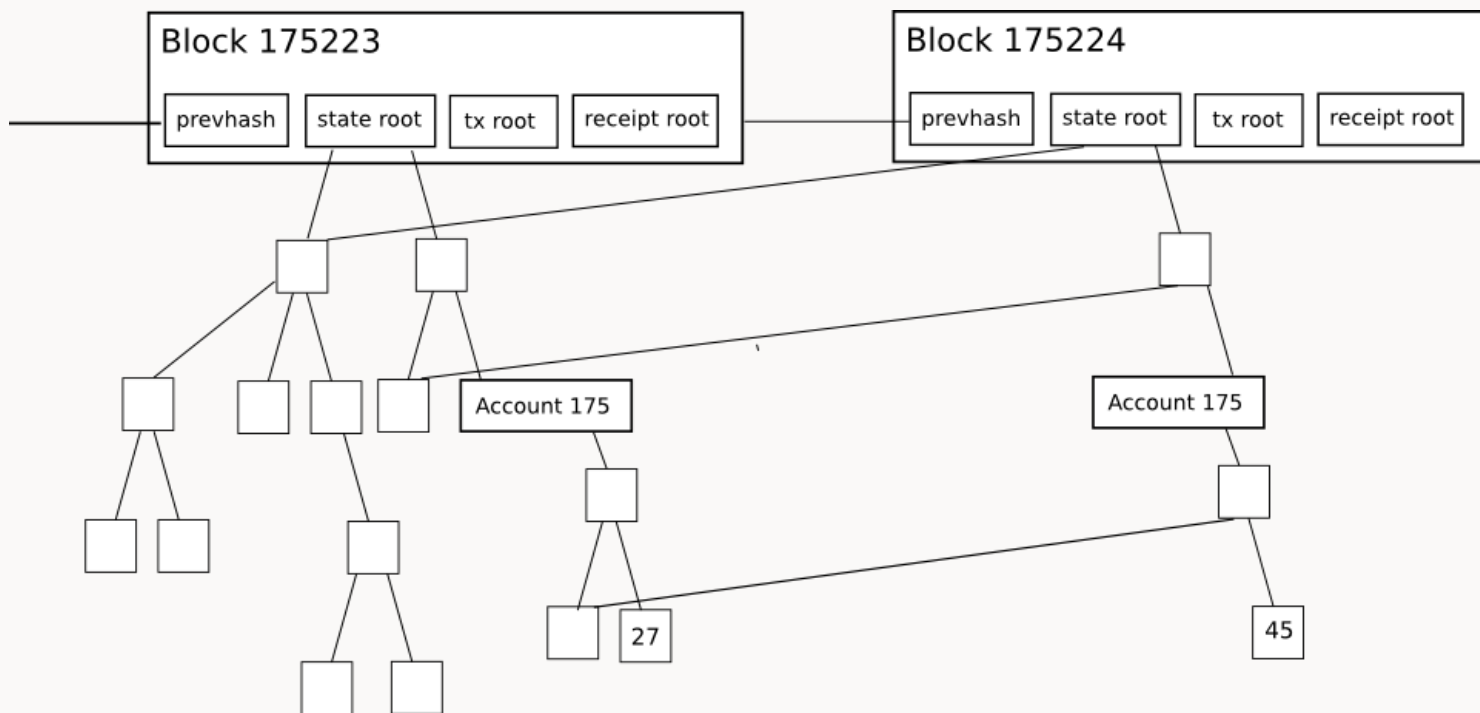
What is Ethereum ?

- State Transition System
- Ethereum State: ID+Balance+Storage+Nonce
- Merkle-Patricia trie
- Block Validation



What is Ethereum ?

- State Transition System
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What is Ethereum ?

- State Transition System
- Ethereum State: ID+Balance+Storage+Nonce
- Merkle-Patricia trie
- Block Validation
- Consensus
 - Ethereum 1.0 : PoW (until 2022)
 - Ethereum 2.0 : PoS

Ethereum 1.0 PoW

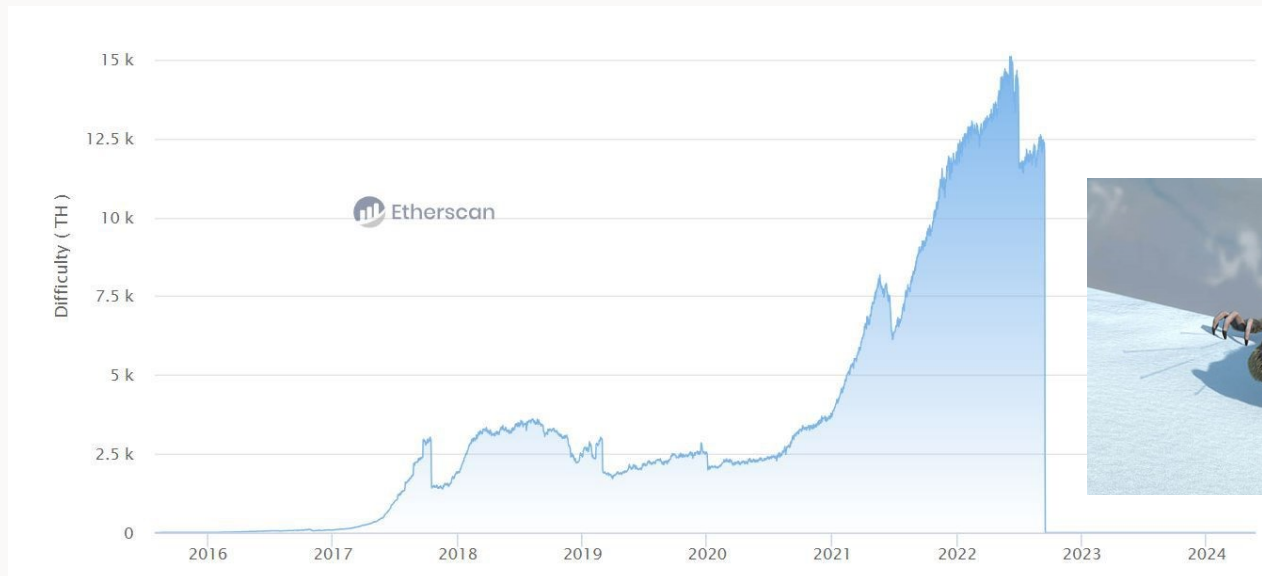
- Use Keccak-256
 - Used in SHA-3
 - mostly use GPU



Ethereum 1.0 PoW

- Use Keccak-256
- Exponential difficulty adjustment
 - ice-age

$$D(H) \equiv \begin{cases} 2^{34} & \text{if } H_i = 0 \\ \max(D_{\min}, P(H)_{H_d} + x \times \varsigma_2 + \epsilon) & \text{otherwise} \end{cases}$$



Ethereum 1.0 PoW

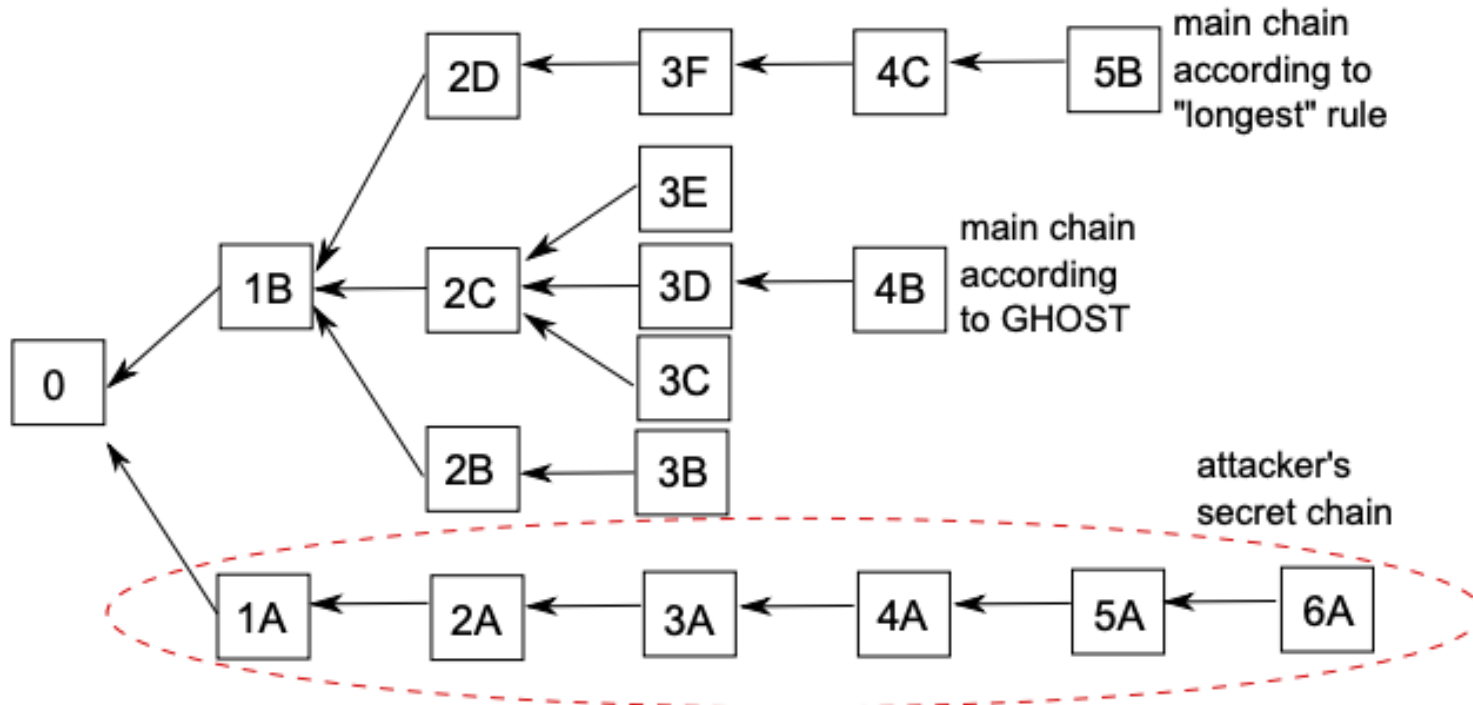
- Use Keccak-256
- Exponential difficulty adjustment
- Block reward
 - adjusted by versioning

$$R_{\text{block}} = 10^{18} \times \begin{cases} 5 & \text{if } H_i < F_{\text{Byzantium}} \\ 3 & \text{if } F_{\text{Byzantium}} \leq H_i < F_{\text{Constantinople}} \\ 2 & \text{if } H_i \geq F_{\text{Constantinople}} \end{cases}$$



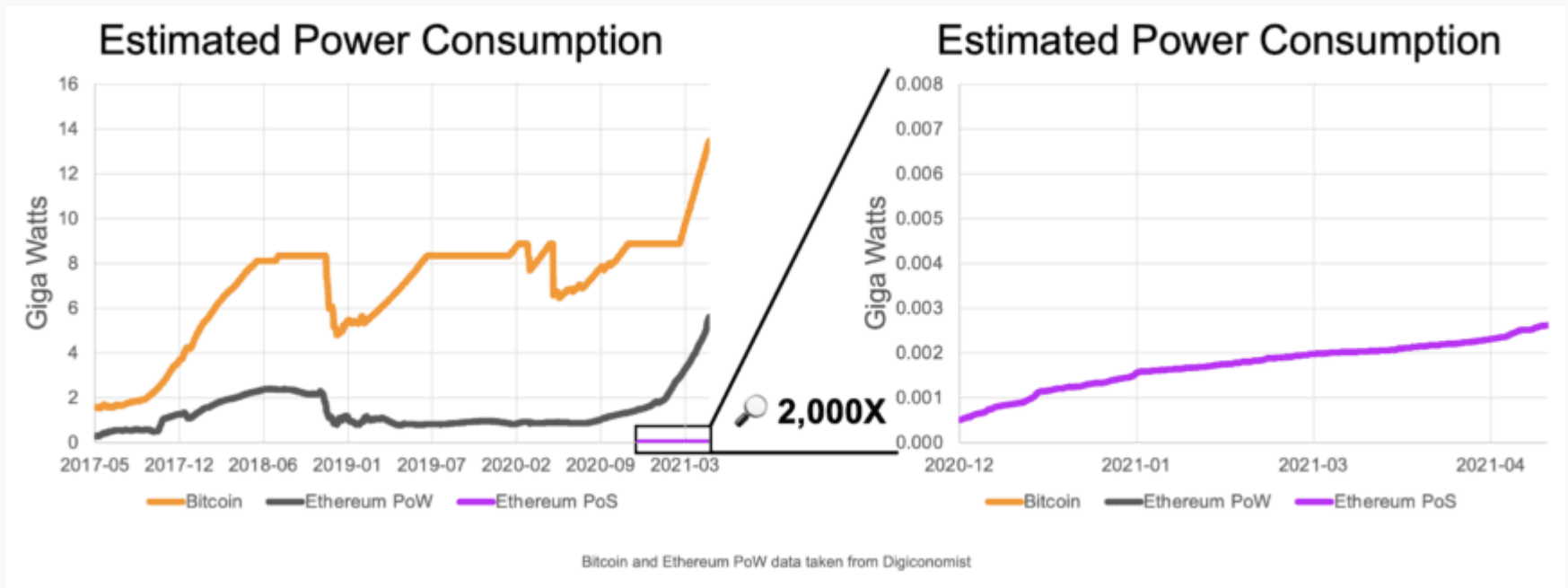
Ethereum 1.0 PoW

- Modified GHOST (Inclusive Protocol)
 - Give 87.5% of base reward to ommer
 - Only up to $2 \leq k \leq 7$ generations



Ethereum 2.0 PoS

- Why ?
 - power consumption of PoW
 - uncle mining attack
 - envisioned from start

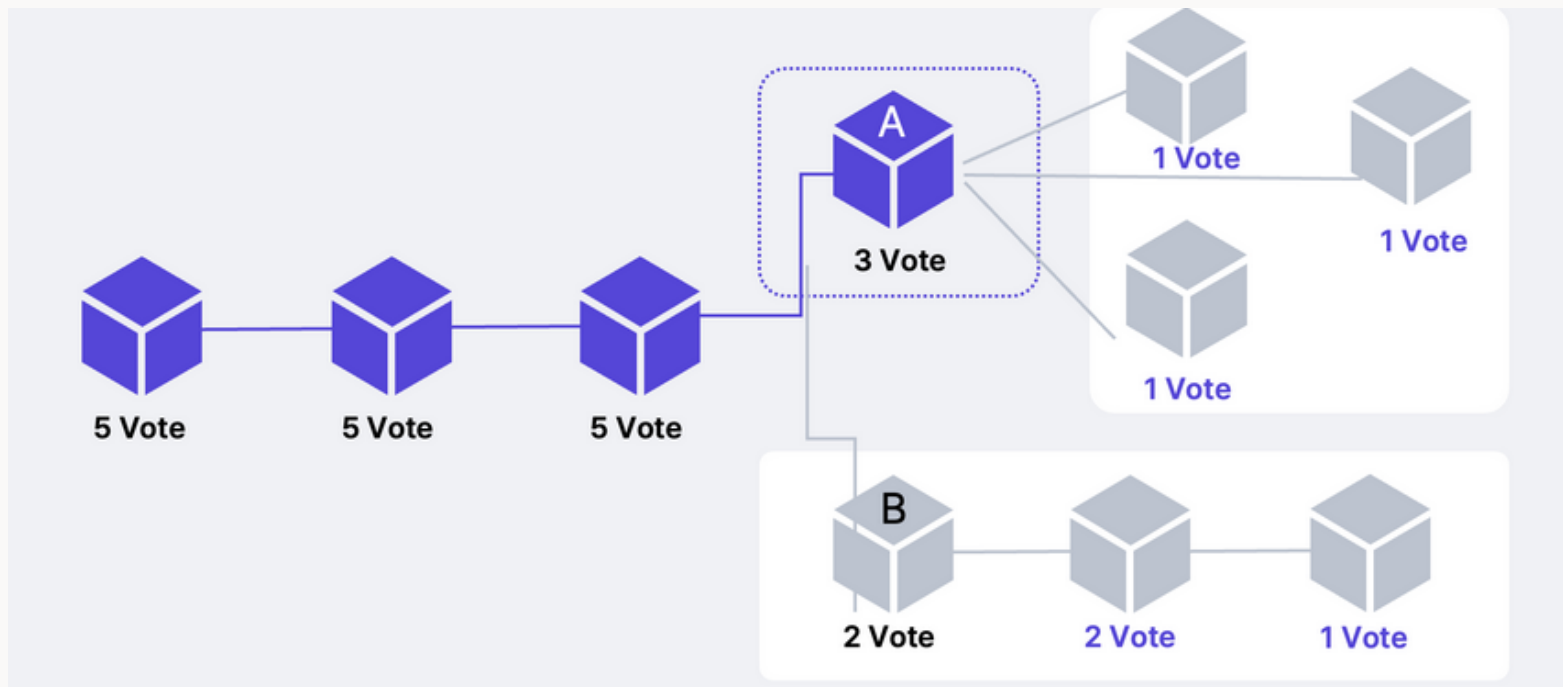


Ethereum 2.0 PoS

- Why ?
- Tested in Beacon Chain in 2020
 - Fixed 32 ETH stake
- Gasper Protocol
 - propose with LMD-GHOST
 - vote with Casper-FFG

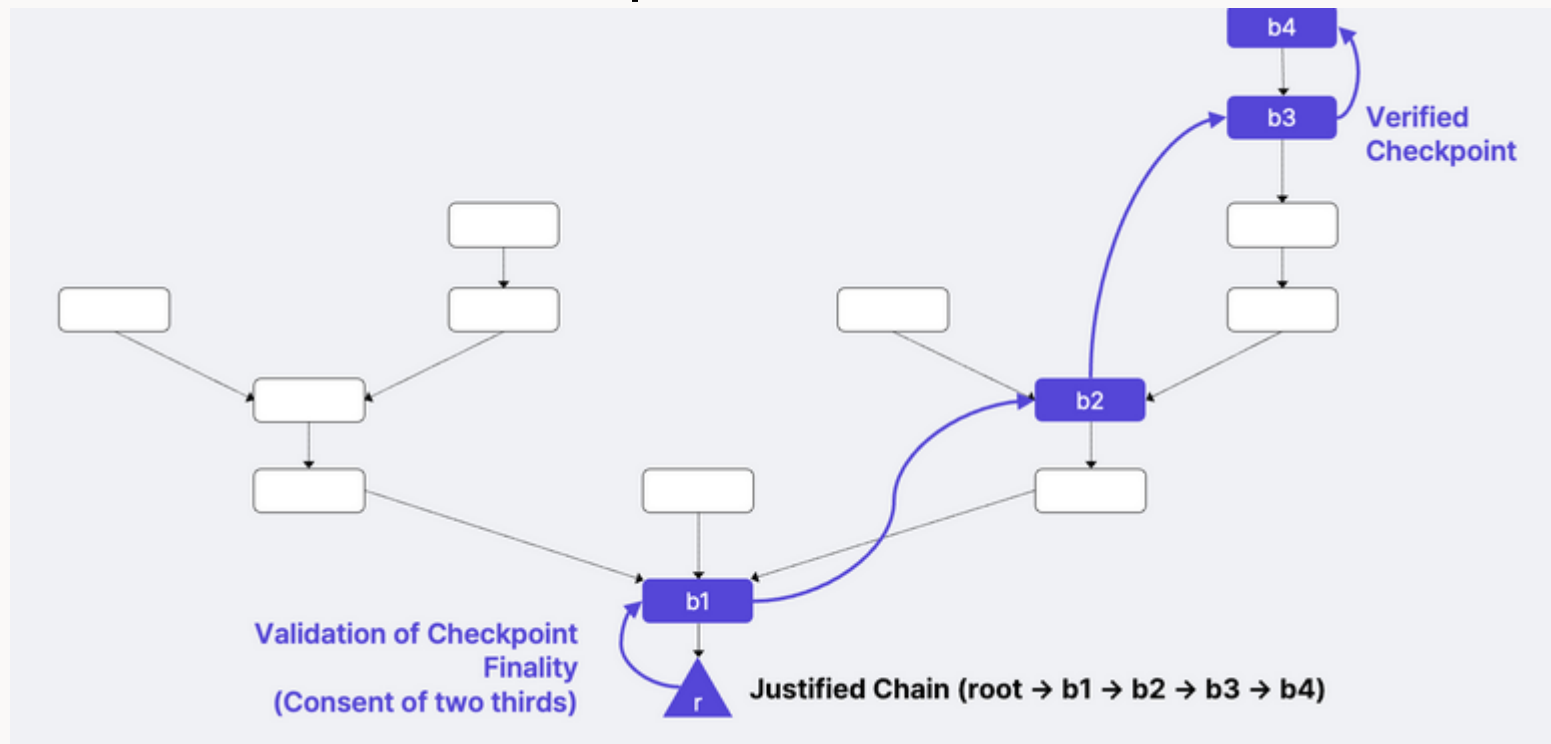
Ethereum 2.0 PoS

- LMD-GHOST
 - Latest Message Driven GHOST
 - 12 second between new blocks



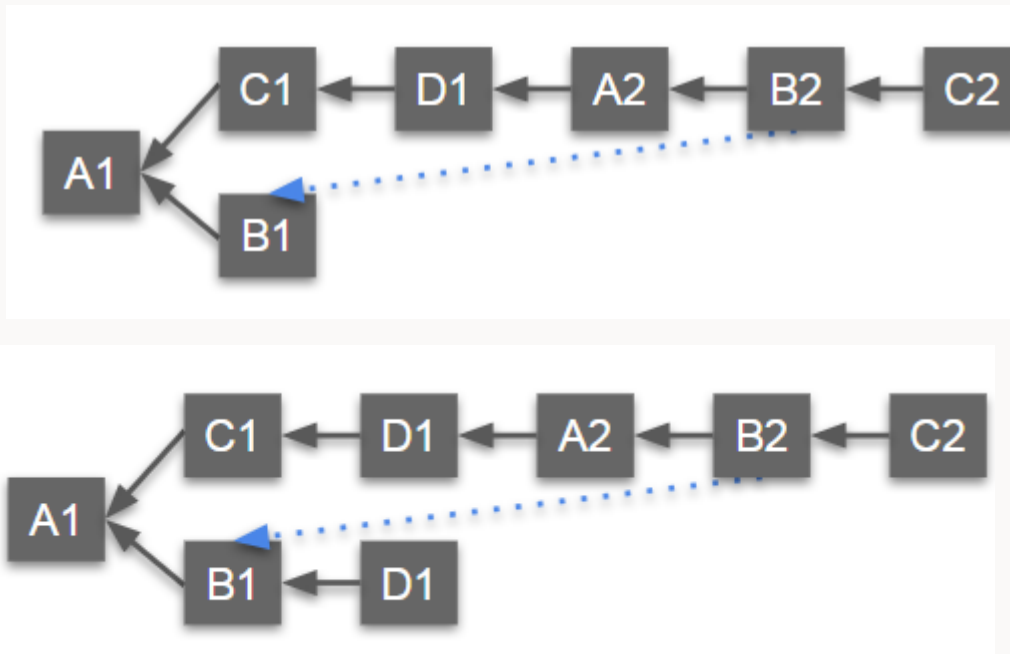
Ethereum 2.0 PoS

- Casper-FFG
 - Friendly Finality Gadget
 - vote once every epoch = 6.4 minutes
 - finalize a checkpoint with $>2/3$ votes



Ethereum 2.0 PoS

- Penalty
 - submitting an invalid block
 - not submitting a block
 - multi vote:



Fee

- Value
 - ETH = $\mathbb{D}$

Multiplier	Name
10^0	Wei
10^9	Gwei
10^{12}	Szabo
10^{15}	Finney
10^{18}	Ether

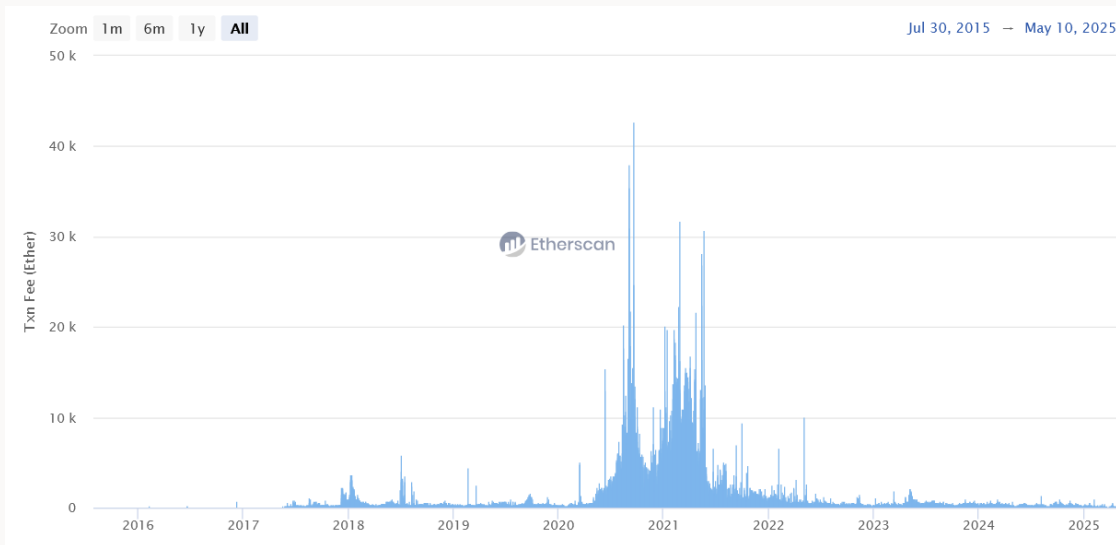
Fee

- Value
- Gas
 - unit of computation and data
 - **gasLimit**: max gas used for a tx
 - **gasPrice**: wei per gas

Gas option	Time	Max fee	
 Low	2 - 3 min	\$3.08 USD	
 Market	30 sec	\$3.87 USD	
 Aggressive	15 sec	\$4.05 USD	
 Advanced	15 sec	\$6.21 USD	
Network status			
49 GWEI Base fee	0.9 - 4 GWEI Priority fee	 Stable	

Fee

- Value
- Gas
 - Expensive when network congested



Fee

- Value
- Gas
- Gasless
 - Pay fee indirectly
 - Gas Station Network (retired)
 - Account Abstraction (EIP 4337)

i You do not have enough ETH in your account to pay for transaction fees on Ethereum Mainnet network. [Buy ETH](#) or deposit from another account.

Reject

Confirm



Fee



- Example

Transaction Hash:	0xa381fc7f929d17108a5f93320e6c60c65ac9a488e0731dba39a4c87ea7626e22
Status:	Success
Block:	22457923 2 Block Confirmations
Timestamp:	20 secs ago (May-11-2025 05:09:23 AM UTC)
From:	0x95222290DD7278Aa3Ddd389Cc1E1d165CC4BAfe5 (beaverbuild)
To:	0x73f7b1184B5cD361cC0f7654998953E2a251dd58 (Fee Recipient: 0x73f7...d58)
Value:	0.090703303037926153 ETH (\$230.19)
Transaction Fee:	0.000119327018831514 ETH (\$0.30)
Gas Price:	4.358500213 Gwei (0.000000004358500213 ETH)
Gas Limit & Usage by Txn:	27,378 27,378 (100%)
Gas Fees:	Base: 4.358500213 Gwei Max: 4.358500213 Gwei Max Priority: 0 ETH
Burnt & Txn Savings Fees:	Burnt: 0.000119327018831514 ETH (\$0.30) Txn Savings: 0 ETH (\$0.00)
Other Attributes:	Txn Type: 2 (EIP-1559) Nonce: 2492665 Position In Block: 162
Input Data:	0x

Fee



- Example

[illegible]

End of Part 1

- Preview of Part 2
 - EVM
 - dApp development
 - DeFi attacks